[INCH-POUND]
A-A-52551
December 15, 1995
SUPERSEDING
MIL-S-62119(AT)
10 October 1969

COMMERCIAL ITEM DESCRIPTION

# STARTER-GENERATOR: 300 AMPERE, 28 VOLTS, DIRECT CURRENT

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers the requirements for a 300 ampere (A), 28 volt (V), direct current (dc) starter-generator, referred to herein as "generator". This dual purpose unit is belt driven with open construction and not submersible. The generator is used on military motor vehicles and on other equipment to start the gasoline engine connected to it and to generate dc energy for lighting, battery charging, and other purposes.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, materials shall conform to the manufacturer's specifications/standards. The use of recycled or reclaimed materials is acceptable provided that all requirements of this CID are met (see 3.1).

2.1.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic corrosion, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

FSC 2920

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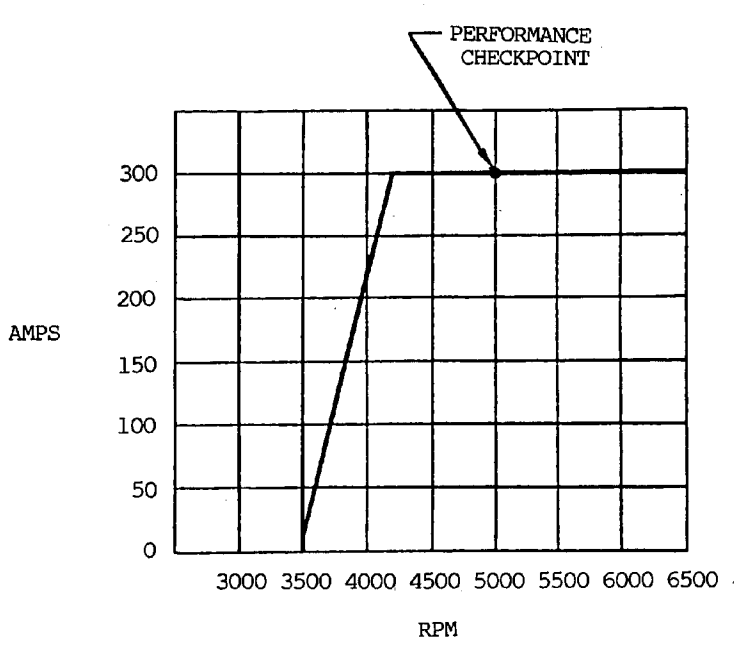
2.2 Design and construction. Generator shall conform to the manufacturer's design and construction with the following exceptions: Generator envelope and interface dimensions and wiring shall conform to Army drawing 12345177. Generator shall be designed to operate with a negatively-grounded electrical circuit. The total weight of the generator shall not exceed 70 pounds.

2.3 Performance.

2.3.1 Output. Generator shall meet the performance requirements in figure 1 when the generator frame is stabilized at the highest attainable temperature, not exceeding 165°F above ambient temperature at 30 volts. Generator shall be considered to be temperature stabilized when operation under constant conditions fails to cause a change in frame temperature in excess of 2°F in any five minute period of any 20 minute test.




NOTES:

1. Generator shall be operated at 28 volts with .75 ohms in series with shunt field terminal "A".
2. Frame shall be stabilized with 140°F max. air inlet temperature.

FIGURE 1. Performance curve.



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2.3.1.1 Maximum speed. When operated at 28 volts, generator shall produce a maximum speed of 3500 revolutions per minute (rpm) at cut-in (see 6.5), a maximum speed of 3600 rpm at 100 A, and a maximum speed of 4200 rpm at 300 A.

2.3.1.2 Starter output. Generator shall develop the minimum torque specified in table I at 16 volts with 1.25 ohms connected external to the shunt field.

TABLE I. Starter output torque.

| Speed, rpm | Current, ampere | Minimum torque, pound-feet |
|---|---|---|
| 1600 | 300 | 12 |
| 1300 | 500 | 20 |
| 0 | --- | 40* |

*Locked-rotor torque

2.3.2 Endurance. Generator shall deliver rated output of 300 amps at 28 volts for 1,000 hours when operated as detailed in table II. During this time, generator shall require no servicing or replacement of parts. Generator shall subsequently meet the requirements of 2.3.1 at the performance checkpoint in figure 1.

TABLE II. Generator endurance schedule.

| Step | Generator speed, rpm (± 25 rpm) | Generator voltage, volts (± 0.2 volts) | Minimum generator output, amps | Time, hours |
|---|---|---|---|---|
| 1 | 3600 | 28 | 300 | 330 |
| 2 | 4500 | 28 | 300 | 335 |
| 3 | 6000 | 28 | 300 | 335 |

NOTE:

During the last 100 hours of operation at each step, generator shall be operated with armature shaft at an angle of 45 ± 1 degrees above the horizontal for the first 50 hours and 45 ± 1 degrees below the horizontal for the second 50 hours.

2.3.3 Overspeed. The generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, raised commutator bars, or other damage when operated with an open field circuit at a speed of 7150 rpm for one minute. The generator shall subsequently meet the requirements of 2.3.1 at the performance checkpoint in figure 1.

2.3.4 Temperature.



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2.3.4.1 High temperature. Generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, raised commutator bars or other damage when operated at average speed, rated output, and ambient temperature of 160° ± 5°F for 4 hours. Generator frame temperature shall not exceed 320°F. Prior to operation, generator shall be stored for 24 hours in an ambient temperature of 160° ± 5°F. Generator shall subsequently meet the requirements of 2.3.1 at the performance checkpoint in figure 1.

2.3.4.2 Low temperature. Generator shall meet the requirements in 2.3.4.1 except that the ambient temperature shall be -65° ± 5°F.

2.3.5 Dielectric strength. Insulation shall evidence no loosening, cracking, charring, burning, smoking, or reduction in dielectric strength when subjected to a ground test of 500 volts root-mean-square (rms) at a frequency of 60 Hertz (Hz), applied for one minute. The potential shall be applied to the armature and field terminals with the ground brush raised off the commutator and with the field ground terminal disconnected.

2.3.6 Shock resistance. Generator shall evidence no breakage, loosening of parts, abrasion, or other physical damage when subjected to three shock pulses conforming to figure 2, applied in both directions along each of three orthogonal axes. Generator shall subsequently meet the requirements of 2.3.1 at the performance checkpoint in figure 1.

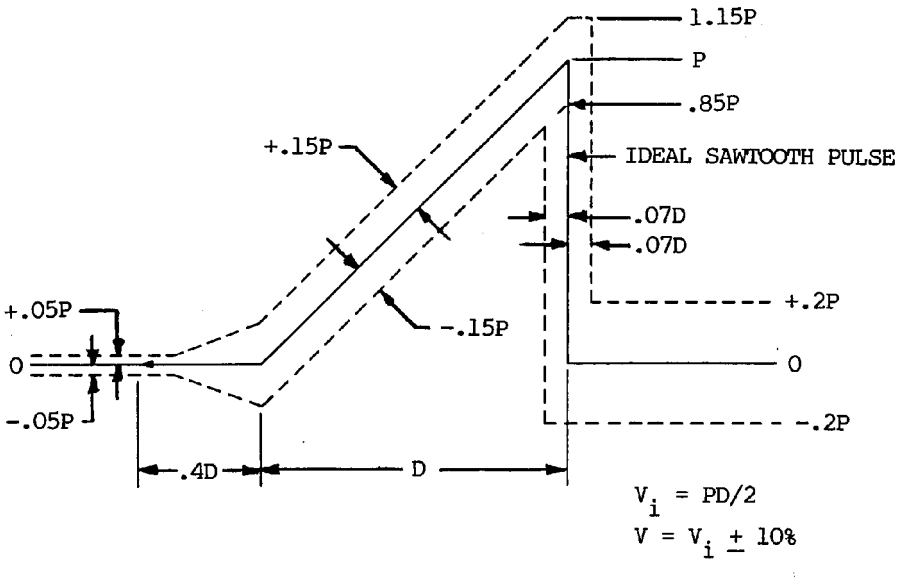


FIGURE 2. Sawtooth shock pulse configuration.



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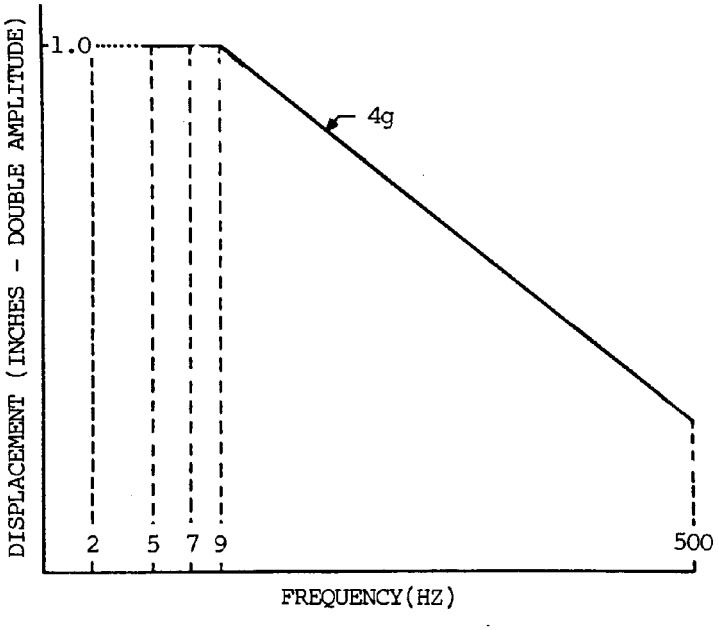
NOTES:

1. Minimum peak value (P) shall be 40 gravity units.
2. Nominal pulse duration shall be 18 milliseconds.
3. The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi $\pm$ 0.1 Vi, where Vi is the velocity change associated with the ideal pulse which equals 0.5 DP. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 2. Sawtooth shock pulse configuration - Continued.

2.3.7 Vibration. Generator shall evidence no breakage, loosening of parts, abrasion, or other physical damage when subjected to a vibration for 2.5 hours along each axis. The vibration frequency shall be varied logarithmically from 5 Hz to 500 Hz and back to 5 Hz in 15 minutes and the vibration amplitude shall be in accordance with figure 3. Generator shall also withstand a vibration at the most severe resonant frequency along each axis for 30 minutes.




NOTE: Curve shall be extended to 2 Hz when resonances below 5 Hz are expected.

FIGURE 3. Vibration curve.



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2.3.8 Electromagnetic interference. Narrowband electromagnetic emissions shall conform to SAE J1816. Broadband electromagnetic emissions shall conform to SAE J551 in the frequency range of 30 MHz to 75 MHz with a 120 kHz bandwidth.

2.3.9 Corrosion resistance. Generator shall be subjected to salt spray testing in accordance with ASTM B117 for 96 hours. At conclusion of the test, the generator shall meet the requirements of 2.3.1 at the performance checkpoint in figure 1, with no impairment of fit, nor Stage 1 corrosion over more than 5% of the area of any component, nor any corrosion past Stage 2.

Corrosion is classified into these two stages:

Stage 1. Red, black and/or white corrosion products on surfaces accompanied by minor etching and/or pitting. Base metal is sound.

Stage 2. Powdered, granular or scaled condition resulting in erosion of material from surface. Base metal is sound with loss of no more than 5%, to a maximum of 0.005 inch reduction of thickness at point of deepest corrosion.

2.3.10 Fungus resistance. Generator shall evidence no microbial growth that will adversely affect performance and shall subsequently meet the requirements of 2.3.1 at the performance checkpoint in figure 1 after exposure to conditions favorable to fungus growth for 90 days.

2.4 Identification marking. Unless otherwise specified (see 6.2), identification marking shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code and part number, and the part identification number (PIN) AA52551-1.

2.5 Surface finish. All exterior surfaces of the generator, except the electrical connections and the extended shaft, shall be surface treated, primed, and painted as specified in the contract or order (see 6.2 and 6.4). No paint shall be applied to mating surfaces.

## 3. REGULATORY REQUIREMENTS.

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

## 4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).



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4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Government documents. Copies of Army drawing 12345177 are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000. Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.1.2 Non-Government documents. Copies of ASTM B117 “Standard Practice for Operating Salt Spray (Fog) Testing Apparatus” are available from American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103-1187. Copies of SAE J551 “Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30 to 1000 MHz)” and SAE J1816 “Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices Narrowband, 10 kHz to 1000 MHz” are available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of the Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. PIN and quantity of generators required.
d. If special marking is required.
e. Surface treatment, priming, painting, and finish color requirements.
f. Applicable level and packaging requirements.



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6.3 Cross-reference. Starter-generators conforming to this CID are interchangeable/substitutable with starter-generators conforming to MIL-S-62119(AT), dated 10 October 1969.

6.4 Finish coat. Starter-generator finish coat should be green 383 in accordance with FED-STD-595.

6.5 Cut-in speed. Cut-in speed is the lowest speed that the starter-generator, with frame temperature stabilized, will generate 28 volts with no external load.

MILITARY INTERESTS:

Custodian:
Army - AT

Review activity:
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 2920-0457)



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